

F. Four Divisors

time limit per test: 10 seconds
memory limit per test: 768 megabytes

If an integer a is divisible by another integer b , then b is called the divisor of a .

For example: 12 has positive 6 divisors. They are 1, 2, 3, 4, 6 and 12.

Let's define a function $D(n)$ — number of integers between 1 and n (inclusive) which has exactly four positive divisors.

Between 1 and 10 only the integers 6, 8 and 10 has exactly four positive divisors. So, $D(10) = 3$.

You are given an integer n . You have to calculate $D(n)$.

Input

The only line contains integer n ($1 \leq n \leq 10^{11}$) — the parameter from the problem statement.

Output

Print the only integer c — the number of integers between 1 and n with exactly four divisors.

Examples

input	Copy
10	
output	Copy
3	

input	Copy
20	
output	Copy
5	

Educational Codeforces Round 12

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
361765700	Feb/07/2026 00:09	Accepted
361763935	Feb/06/2026 23:39	Accepted
322705549	Jun/03/2025 18:17	Accepted
296813999	Dec/16/2024 11:56	Time limit exceeded on test 11
260904311	May/14/2024 12:29	Accepted
240686695	Jan/07/2024 14:49	Accepted
236701937	Dec/10/2023 23:04	Wrong answer on test 13
236701840	Dec/10/2023 23:03	Compilation error
236701637	Dec/10/2023 23:00	Wrong answer on test 13
236701621	Dec/10/2023 23:00	Compilation error

→ Problem tags

data structuresdpmath

number theorysortings two pointers

*2400

No tag edit access

→ Contest materials

Announcement (en)